Study 3 merged dataset

2022-09-14

Contents

Columns in the the "study3dataCleaned.csv" dataset **2**

Loading data and packages **3**

Matching study 3 item numbers to SAPA item numbers **3**
SAPA item numbers . . . 3
Study 3 item numbers . . . 4

Renaming columns in both datasets using SAPA item numbers **5**
CRA item columns . . . 5
CRA item display order columns . . . 5
Renaming remaining columns . . . 6
Demographics . . . 6
Cognitive Reflection Test and Four Factor Imagination Scale . . . 7
"Seen before" columns from session 2 . . . 7

Merging the session 1 and session 2 datasets **8**

Scoring the CRA items from session 1 and 2 **8**

Columns in the the "study3dataCleaned.csv" dataset

- **Columns 1 through 191 are from session 1:**
  - **Timing information and demographics**: columns 1 ("StartDate") through 13 ("gender_selfdescribe")
  - **CRA items**: columns 14 ("q_14003") through 79 ("q_14197")
  - **Attention check questions**: columns 80 ("FAKE_1") and 81 ("FAKE_2")
  - **Cognitive Reflection Task**: columns 82 ("CRT_1") through 91 ("CRT_10")
  - column 92 ("FAKE_3") is an attention check item
  - **Four Factor Imagination Scale**: columns ("FFIS_F1") through 118 ("FFIS_D5")
  - column 119 ("FAKE_4") is an attention check item
  - **Display order for CRA items**: columns 120 ("MultipleChoice_DO_q_14006") through 191 ("MultipleChoice_DO_q_14051")
- **Columns 192 through 362 are from the retest session:**
  - **Timing information and practice**: columns 192 ("StartDate_retest") through 197 ("Practice_retest")
  - **CRA items**: columns 198 ("q_14003_retest") through 296 ("q_14197_retest"). There is one column after each of the first 33 CRA items labeled "Seen_Before.." for questions asking participants: "Did you see this item during the previous session?".
  - **Display order for CRA items**: columns 297 ("MultipleChoice_DO_q_14066_retest") through 362 ("MultipleChoice_DO_q_14051_retest")
- Column 363 ("Retest_sample") indicates whether the person participated in the retest session ("1") or not ("0")
- **Columns 364 through 505 are the scored items:** ("1" for correct answers and "0" for incorrect answers)
  - **Scored CRA items for session 1**: columns 364 ("q_14003_scored") through 429 ("q_14197_scored")
  - **Scored CRA items for session 2**: columns 430 ("q_14003_retest_scored") through 495 ("q_14197_retest_scored")
  - **Scored CRT items**: columns 496 ("CRT_1_scored") through 505 ("CRT_10_scored")

Loading data and packages

- The file `CRA-Prolific_Session1_Raw_data_Sept142022.csv` contains the raw data for study 3 at time point 1 (responses to the CRA problems, the Cognitive Reflection Task, and Four Factor Imagination Scale).
- The file `CRA_Retest_Raw_Viewing_Order_Included.xlsx` contains the raw data from time-point 2, which includes participants' viewing orders for the CRA problems they were administered.
- The file `ICARcraSampleItems-2.xlsx` contains the SAPA item numbers for the CRA problems (q_14003 through q_14197).

```
library(rio)
library(tidyverse)
library(here)
# Session 1 data
Prolific_1_RAW <- import(here::here("Data", "CRA-Prolific_Session1_Raw_data_Sept142022.csv"))
# Session 2 data
Prolific_2_RAW <- import(here::here("Data", "CRA_Retest_Raw_Viewing_Order_Included.xlsx"))
Prolific_2_RAW <- Prolific_2_RAW[-c(2:19),] # Removing entries from us testing the Qualtrics survey
# SAPA item numbers
SAPAlabels <- import(here::here("ICARcraSampleItems-2.xlsx"))
```

Matching study 3 item numbers to SAPA item numbers

Below, I create a dataframe that has—-for each CRA item–the stimuli words, solution word, SAPA item number, and the qualtrics item number. I use this dataframe to rename CRA columns in the qualtrics with the correct SAPA item numbers.

SAPA item numbers

```
items1 <- SAPAlabels %>%
  # Selecting only CRA items undergoing validation
  filter(validation1 == "1") %>%
  # Selecting the SAPA item numbers, 3 stimuli words, and solution word
  select("item num", prompt1, prompt2, prompt3, stem) %>%
  rename(SAPA_item_number="item num") %>%
  # putting "q_" in front of SAPA item numbers (e.g., 14003 becomes q_14003)
  mutate(SAPA_item_number = paste0("q_", SAPA_item_number, sep = "")) %>%
  # Creating a variable with the three stimuli words merged together (as shown in qualtrics)
  mutate(prompt = paste0(prompt1, prompt2, prompt3, sep="")) %>%
  relocate(prompt, .after = SAPA_item_number) %>%
  # Creating another prompt variable to reflect the prompts shown in Qualtrics
  mutate(Qualtrics_prompt = prompt) %>%
  relocate(Qualtrics_prompt, .after = prompt)

# Editing one of the "Qualtrics_prompt" columns. This was necessary because I typed one of the stimuli

items1[9,3] = "linerockmay"
```

Table 1: SAPA item numbers and item content for each of the CRA problems (first 5 rows)

SAPA_item_number	prompt	Qualtrics_prompt	prompt1	prompt2	prompt3	stem
q_14003	bedsheetwide	bedsheetwide	bed	sheet	wide	spread
q_14004	warthoroughfan	warthoroughfan	war	thorough	fan	fare
q_14006	soilmountaintable	soilmountaintable	soil	mountain	table	top
q_14007	junkgravestick	junkgravestick	junk	grave	stick	yard
q_14008	stonewordhole	stonewordhole	stone	word	hole	key

Study 3 item numbers

The Qualtrics item numbers and content for each item (in row 1 for the session 1 and retest datasets) are added to the dataframe with SAPA item numbers.

```
# Session 1 data first
Study3_session1_items <- data.frame(t(Prolific_1_RAW[1,])) %>%
  mutate(Session1_item_number = rownames(.)) %>%
  rename(Qualtrics_prompt = X1) %>%
  # Merging using the "Qualtrics_prompt" column
  full_join(items1, by = "Qualtrics_prompt") %>%
  relocate(Session1_item_number, .after = "SAPA_item_number")
```

Table 2: SAPA item numbers, Session 1 item numbers, and item content for the CRA problems

	Qualtrics_prompt	SAPA_item_number	Session1_item_number	prompt	prompt1	prompt2	prompt3	stem
27	bedsheetwide	q_14003	Q818	bedsheetwide	bed	sheet	wide	spread
28	warthoroughfan	q_14004	Q822	warthoroughfan	war	thorough	fan	fare
29	soilmountaintable	q_14006	Q826	soilmountaintable	soil	mountain	table	top
30	junkgravestick	q_14007	Q830	junkgravestick	junk	grave	stick	yard
31	stonewordhole	q_14008	Q834	stonewordhole	stone	word	hole	key

Now adding the item numbers from the retest dataset

```
# Retest dataset
Study3_items <- data.frame(t(Prolific_2_RAW[1,])) %>%
  mutate(Session2_item_number = rownames(.)) %>%
  rename(Qualtrics_prompt = X1) %>%
  full_join(Study3_session1_items, by = "Qualtrics_prompt") %>%
  relocate(Session2_item_number, .after = "Session1_item_number")
```

Table 3: SAPA item numbers, Sessions 1 and 2 item numbers, and item content for the CRA problems

	Qualtrics_prompt	SAPA_item_number	Session1_item_number	Session2_item_number	prompt	prompt1	prompt2	prompt3
27	Have you seen this problem before?	NA	NA	Q269	NA	NA	NA	NA
28	stonewordhole	q_14008	Q834	Q834	stonewordhole	stone	word	hole
29	Have you seen this problem before?	NA	NA	Q270	NA	NA	NA	NA
30	cakeboardtea	q_14009	Q838	Q838	cakeboardtea	cake	board	tea
31	Have you seen this problem before?	NA	NA	Q271	NA	NA	NA	NA

Checking to see whether any of the item numbers for the CRA items differ between sessions 1 and 2 (they shouldn't, since the Qualtrics survey for session 1 was copied and redistributed for session 2).

```
Study3_items %>%
  filter(!is.na(SAPA_item_number)) %>%
  filter(Session1_item_number != Session2_item_number)
```

```
##   Qualtrics_prompt SAPA_item_number Session1_item_number Session2_item_number
## 1 bitterheartbread          q_14025               Q870.1                 Q870
```

```
##              prompt prompt1 prompt2 prompt3  stem
## 1 bitterheartbread  bitter   heart   bread sweet
```

Just one difference: "Q870" in session 2 is "Q870.1" in session 1. It looks like the session 1 column was labeled this way because one of the demographic questions about English proficiency already had the label "Q870". Below, I change the label for the English proficiency question in study 1 and use the "Q870" label for the CRA item in both sessions 1 and 2.

```
# First, filtering any unencessary rows
Study3_items_renaming <- Study3_items %>%
  filter(!is.na(SAPA_item_number) & !is.na(Session1_item_number)) %>%
  select(SAPA_item_number, Session1_item_number, Session2_item_number,
  prompt, Qualtrics_prompt)

Study3_items_renaming[14,2] = "Q870"
colnames(Prolific_1_RAW)[21] = 'english'
```

Renaming columns in both datasets using SAPA item numbers

CRA item columns

```
# Session 1
oldnames1 = as.character(Study3_items_renaming$Session1_item_number)
newnames1 = as.character(Study3_items_renaming$SAPA_item_number)

Prolific_1_renamed <- Prolific_1_RAW %>%
  rename_with(~ newnames1[which(oldnames1 == .x)], .cols = oldnames1)

# Session 2
oldnames2 = as.character(Study3_items_renaming$Session2_item_number)
newnames2 = paste0(as.character(Study3_items_renaming$SAPA_item_number), "_retest")

Prolific_2_renamed<- Prolific_2_RAW %>%
  rename_with(~ newnames2[which(oldnames2 == .x)], .cols = oldnames2)
```

For clarification, I just replaced the Qualtrics item numbers (oldnames below) with the matching SAPA item numbers (newnames below)

```
oldnames1[1:5]
```

```
## [1] "Q818" "Q822" "Q826" "Q830" "Q834"
```

```
newnames1[1:5]
```

```
## [1] "q_14003" "q_14004" "q_14006" "q_14007" "q_14008"
```

CRA item display order columns

I use the same approach to rename the display order columns for the CRA items.

```
# Session 2
Study3_items_renaming <- Prolific_2_renamed %>%
  select(contains("MultipleChoice_DO_")) %>%
  colnames() %>%
  as.data.frame() %>%
  rename(Display_order = ".") %>%
  mutate(Session2_item_number = gsub("MultipleChoice_DO_", "", Display_order)) %>%
```

```
  inner_join(Study3_items_renaming, by = "Session2_item_number") %>%
  mutate(Display_order_new = paste0("MultipleChoice_DO_", SAPA_item_number, sep=""))

oldnames2 = as.character(Study3_items_renaming$Display_order)
newnames2 = paste0(as.character(Study3_items_renaming$Display_order_new), "_retest")

Prolific_2_renamed <- Prolific_2_renamed %>%
  rename_with(~ newnames2[which(oldnames2 == .x)], .cols = oldnames2)

# Session 1
Study3_items_renaming <- Prolific_1_renamed %>%
  select(contains("MultipleChoice_DO_")) %>%
  colnames() %>%
  as.data.frame() %>%
  rename(Display_order1 = ".") %>%
  mutate(Session1_item_number = gsub("MultipleChoice_DO_", "", Display_order1)) %>%
  inner_join(Study3_items_renaming, by = "Session1_item_number")


oldnames1 = as.character(Study3_items_renaming$Display_order1)
newnames1 = as.character(Study3_items_renaming$Display_order_new)

Prolific_1_renamed <- Prolific_1_renamed %>%
  rename_with(~ newnames1[which(oldnames1 == .x)], .cols = oldnames1)
```

```
oldnames1[1:5]
```

```
## [1] "MultipleChoice_DO_Q826"  "MultipleChoice_DO_Q1018"
## [3] "MultipleChoice_DO_Q906"  "MultipleChoice_DO_Q822"
## [5] "MultipleChoice_DO_Q866"
```

```
newnames1[1:5]
```

```
## [1] "MultipleChoice_DO_q_14006" "MultipleChoice_DO_q_14159"
## [3] "MultipleChoice_DO_q_14066" "MultipleChoice_DO_q_14004"
## [5] "MultipleChoice_DO_q_14024"
```

Renaming remaining columns

Demographics

```
oldnames2 = as.character(colnames(Prolific_2_renamed)[1:17])
newnames2 = paste0(as.character(colnames(Prolific_2_renamed)[1:17]), "_retest")

Prolific_2_renamed <- Prolific_2_renamed %>%
  rename_with(~ newnames2[which(oldnames2 == .x)], .cols = oldnames2)

Prolific_1_renamed <- Prolific_1_renamed %>%
  rename(Enter_Prolific_ID = Q293,
         Practice = Q2,
         english_native = Q869,
         education = Q871,
         income = Q872,
         age = Q291,
         gender = Q292,
```

```
         gender_selfdescribe = Q292_5_TEXT)

Prolific_2_renamed <- Prolific_2_renamed %>%
  rename(Practice_retest = Q2,
         Enter_Prolific_ID_retest = Q338)
```

Cognitive Reflection Test and Four Factor Imagination Scale

```
# New column names for each scale and subscale
CRT_cols = as.character(paste("CRT", c(1:10), sep = "_"))
FFIS_F_cols = as.character(paste("FFIS_F", c(1:9), sep = ""))
FFIS_C_cols = as.character(paste("FFIS_C", c(1:5), sep = ""))
FFIS_EV_cols = as.character(paste("FFIS_EV", c(1:7), sep = ""))
FFIS_D_cols = as.character(paste("FFIS_D", c(1:5), sep = ""))

# Qualtrics column names to be changed for each scale
# CRT
old_CRT_cols = Prolific_1_renamed %>%
  select(Q262:Q286) %>%
  names()
# FFIS
# Frequency
old_FFISF_cols = Prolific_1_renamed %>%
  select(Q246:Q255) %>%
  names()
# Complexity
old_FFISC_cols = Prolific_1_renamed %>%
  select(Q256:Q260) %>%
  names()
# Emotional Valence
old_FFISEV_cols = Prolific_1_renamed %>%
  select(Q267:Q273) %>%
  names()
# Directedness
old_FFISD_cols = Prolific_1_renamed %>%
  select(Q274:Q278) %>%
  names()

## Renaming
Prolific_1_renamed <- Prolific_1_renamed %>%
  rename_with(~ CRT_cols[which(old_CRT_cols == .x)], .cols = old_CRT_cols) %>%
  rename_with(~ FFIS_F_cols[which(old_FFISF_cols == .x)], .cols = old_FFISF_cols) %>%
  rename_with(~ FFIS_C_cols[which(old_FFISC_cols == .x)], .cols = old_FFISC_cols) %>%
  rename_with(~ FFIS_EV_cols[which(old_FFISEV_cols == .x)], .cols = old_FFISEV_cols) %>%
  rename_with(~ FFIS_D_cols[which(old_FFISD_cols == .x)], .cols = old_FFISD_cols)
```

"Seen before" columns from session 2

```
# Session 2
old_seen_cols <- Prolific_2_renamed %>%
  select(q_14003_retest:Q299, -contains("q_")) %>%
  names()
```

```
# Seen cols
seen_cols = as.character(paste("Seen_Before", c(1:33), sep = "_"))
Prolific_2_renamed <- Prolific_2_renamed %>%
  rename_with(~ seen_cols[which(old_seen_cols == .x)], .cols = old_seen_cols)

# Display order for seen cols
old_seen_cols_DO <- Prolific_2_renamed %>%
  select(contains(old_seen_cols)) %>%
  names()

seen_cols_DO <- paste(old_seen_cols_DO, seen_cols, sep = "_")

Prolific_2_renamed <- Prolific_2_renamed %>%
  rename_with(~ seen_cols_DO[which(old_seen_cols_DO == .x)],
  .cols = old_seen_cols_DO)
```

Creating a variable indicating whether participants were included in session 2 (their Prolific ID is in the retest dataset)

```
Prolific_2_renamed <- Prolific_2_renamed %>%
  mutate(Retest_sample = "1")
```

Merging the session 1 and session 2 datasets

```
Prolific_data <- full_join(Prolific_1_renamed, Prolific_2_renamed,
  by = "PROLIFIC_PID") %>%
  # Removing display order for FFIS and CRT scales
  select(-contains("FFIS_DO"), -contains("CRT_DO"))
```

Scoring the CRA items from session 1 and 2

Getting separate dataframes with just the session 1 and session 2 items to score separately.

```
CRA_session1 <- Prolific_data %>%
  select(contains("q_"), -contains("_retest"), -contains("Multiple"))

CRA_session2 <-Prolific_data %>%
  select(contains("q_") & contains("_retest"), -contains("Multiple"))
```

Recoding -99s in both datasets as "NA". This ensures that no skipped items are scored as incorrect (they should not have been able to skip items. -99 values likely indicate some sort of technical error).

```
CRA_session2 <- psych::scrub(CRA_session2,isvalue="-99")
CRA_session1 <- psych::scrub(CRA_session1,isvalue="-99")
```

The function 'Score_CRA_MC' below scores the CRA items using a key with the solutions (the first letter of each solution word).

```
Solutions <- import(here("ICARcraSampleItems-2.xlsx")) # Getting solutions
Solutions <- Solutions %>% filter(validation1 == "1") # Selecting only CRA problems given in this study
Solutions <- Solutions[1:67,] # Selecting only problems given in this study
Solutions <- Solutions[-c(35),] # Removing "Pine/Crab/Sauce" (the example item, which was not given)
Solutions <- Solutions$stem # Selecting only the solutions to each problem (the "stem" word)
Solutions <- str_sub(Solutions, 1, 1) # Multiple choice items have only the first letter as solution
```

```
Score_CRA_MC <- function(word_list, dataset) {

   frame <- matrix(nrow=nrow(dataset), ncol=ncol(dataset))

   for(i in seq_along(dataset)) {

      frame[,i] <-  ifelse(dataset[,i] == word_list[i], 1, 0)
      rownames(frame) <- seq(1:nrow(dataset))
      colnames(frame) <- colnames(dataset)
      colnames(frame) <- paste(colnames(frame), "scored", sep = "_")


   }

      frame <- as.data.frame(frame)
      #dataset <- cbind(dataset, frame)
      #return(dataset)
      return(frame)

}
```

Scoring the CRA items from each session:

```
## Session 1 scoring:
Session1_CRA_Scored <- Score_CRA_MC(word_list = Solutions, dataset = CRA_session1)
Prolific_data <- cbind(Prolific_data, Session1_CRA_Scored)
## Session 2 scoring
Session2_CRA_Scored <- Score_CRA_MC(word_list = Solutions, dataset = CRA_session2)
Prolific_data <- cbind(Prolific_data, Session2_CRA_Scored)
```